LESSON 5.9

Area of Regular Polygons

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a regular polygon by decomposing it into triangles.

WHY THIS MATTERS Your architecture firm is designing a hexagonal skylight for the lobby of a new museum. The glass fabricator needs to know the total area so they can cut the correct amount of glass. Each side of the hexagon is 6 feet, and the height of each triangle formed from the center is 5.2 feet. The height is rounded to the nearest tenth of a foot, so the area is approximate.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Regular Polygon *Spanish: Polígono regular*	A closed flat shape made of straight sides, where every side is the same length and every angle is the same size.	A stop sign is a regular octagon — all 8 sides are the same length and all 8 angles are equal
Decompose *Spanish: Descomponer*	To break a shape into smaller, simpler shapes.	Draw lines from the center of a hexagon to each corner → you get 6 equal triangles you can find the area of
Triangle *Spanish: Triángulo*	A shape with three sides.	Each triangle inside a decomposed hexagon has a base = one side of the hexagon and height = distance from center to side
Composite *Spanish: Compuesto*	Made by putting two or more simple shapes together.	6 triangles, each with area 15 sq ft, combine to form a hexagon with total area = 6 × 15 = 90 sq ft
Formula *Spanish: Fórmula*	A math rule written with symbols.	For a regular hexagon: Total Area = 6 × (½ × b × h), where b = side length and h = height of each triangle

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

My hexagon has 6 equal sides. Each side is 6 feet, and each center triangle has a height of 5.2 feet. Use the rounded height to estimate the area.

1. First I find one triangle: A = ½ × base × height = ½ × 6 × 5.2.
2. I multiply: 6 × 5.2 = 31.2, then ½ × 31.2 = 15.6 square feet.
3. A hexagon has 6 sides, so it has 6 equal triangles.
4. I multiply: 6 × 15.6 = 93.6. So the total area is 93.6 square feet.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a smaller regular pentagon: each of its 5 triangles has base 4 cm and height 3 cm. Use the rounded height to estimate the area.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The total area is _____ because one triangle is _____ and there are _____ triangles.

Word bank Regular Polygon | Decompose | Triangle | Composite | Formula | triangle | half | base | height | multiply | total | number of sides

WATCH OUT!

A common mistake

A common mistake in Area of Regular Polygons is stopping after finding the area of ONE triangle and reporting that as the total area

LESSON 5.9

Area of Regular Polygons

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS Your architecture firm is designing a hexagonal skylight for the lobby of a new museum. The glass fabricator needs to know the total area so they can cut the correct amount of glass. Each side of the hexagon is 6 feet, and the height of each triangle formed from the center is 5.2 feet. The height is rounded to the nearest tenth of a foot, so the area is approximate.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

My hexagon has 6 equal sides. Each side is 6 feet, and each center triangle has a height of 5.2 feet. Use the rounded height to estimate the area.

1. First I find one triangle: A = ½ × base × height = ½ × 6 × 5.2.
2. I multiply: 6 × 5.2 = 31.2, then ½ × 31.2 = 15.6 square feet.
3. A hexagon has 6 sides, so it has 6 equal triangles.
4. I multiply: 6 × 15.6 = 93.6. So the total area is 93.6 square feet.

1 **Circle the letter of the best answer. Show how you know.**

A regular hexagon is divided into 6 equal triangles from the center. Each triangle has a base of 6 ft and a height of 5.2 ft. What is the area of one triangle? Use the given heights; nonsquare heights are rounded, so their areas are estimates.

(A) 11.2 sq ft
(B) 15.6 sq ft
(C) 15.6 ft
(D) 31.2 sq ft

STEPS TO TRY

1. Focus on just ONE of the 6 triangles: its base is 6 ft and its height is 5.2 ft.
2. One triangle uses A = ½ × b × h — don't multiply by 6 yet; that step is for the whole hexagon.
3. Compute ½ × 6 × 5.2: take half of 6 first, then multiply by 5.2. Label it in sq ft.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

2 **Circle the letter of the best answer. Show how you know.**

A regular hexagonal skylight has 6 central triangles, each with base 6 ft and rounded height 5.2 ft. Using those measurements, what is the approximate total area?

(A) 15.6 sq ft
(B) 62.4 sq ft
(C) 93.6 sq ft
(D) 187.2 sq ft

STEPS TO TRY

1. Find one central triangle's area: ½ × 6 × 5.2 = 15.6 sq ft using the rounded height.
2. A regular hexagon splits into 6 identical triangles, so the total is 6 copies of one triangle's area.
3. Multiply 15.6 × 6. (Tip: do 15 × 6, then 0.6 × 6, and combine.)

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

A regular hexagon is split into 6 triangles, each with base 4 cm and height 3.5 cm. What is the total area? Use the given heights; nonsquare heights are rounded, so their areas are estimates.

(A) 14 sq cm
(B) 21 sq cm
(C) 42 sq cm
(D) 84 sq cm

STEPS TO TRY

1. Two stages hide here: first one triangle (base 4 cm, height 3.5 cm), then all 6 of them.
2. Use ½ × 4 × 3.5 for one triangle, then multiply that result by 6 for the hexagon.
3. Compute ½ × 4 = 2, then 2 × 3.5 for one triangle. Now scale up by the 6 triangles.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

How many triangles can you make from the center of a regular pentagon?

(A) 3
(B) 5
(C) 6
(D) 10

STEPS TO TRY

1. 'Penta' is a clue about how many sides a pentagon has.
2. Drawing from the center to each corner makes exactly one triangle per side of the polygon.
3. Count the pentagon's sides — that count equals the number of triangles.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

The hexagon and the octagon both have triangles, but the totals are different. What two things decide the total area of a regular polygon?

SENTENCE STARTER The total area is ___ because one triangle is ___ and there are ___ triangles.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help